

Online Supplement for “A simheuristic for the cross-docking scheduling problem under uncertain truck arrival times”

To support the main manuscript, this supplement provides mathematical proofs (Section OS.1), experimental configurations (Section OS.2), detailed comparison results (Section OS.3), the benchmark MILP model (Section OS.4), sensitivity analyses (Section OS.5), and algorithmic ablation studies (Section OS.6).

OS.1. Proofs

The proof of Proposition 1

Proof. Let s^* denote the optimal start times under the given sequence π , and define $\bar{s}$ such that:

$$\bar{s}_i = \begin{cases} ES_i, & i \in \mathcal{I}, \\ s_i^*, & i \in \mathcal{O}, \end{cases}$$

Let the deterministic objective value under sequence π and schedule s be:

$$DetC(\pi, s) = c \left[\sum_{i \in \mathcal{I}} \max(0, s_i + g_i - e_i) + \sum_{j \in \mathcal{O}} \max(0, s_j + g_j - e_j) \right] + \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{O}} \eta_{ij},$$

where e_i denotes the expected departure time for truck i , and η_{ij} is the penalty cost component involving goods f_{ij} . Since the EST rule always produces $\bar{s}_i \leq s_i^*$ for all $i \in \mathcal{I}$, and $\bar{s}_j = s_j^*$ for all $j \in \mathcal{O}$, it follows that: $DetC(\pi, \bar{s}) \leq DetC(\pi, s^*)$.

Therefore, $\bar{s}$ is also an optimal start time under the fixed sequence π . □

The proof of Theorem 1

Proof. We prove the theorem by induction on the position of trucks in the sequence π_l at a given outbound door $l \in \mathcal{D}^O$. Consider the first truck $j_1 \in \pi_l$, the realized starting time of truck j_1 is

$$\hat{s}_{j_1\omega} = \max\{\hat{a}_{j_1}^\omega, \tau_{j_1}\} = \max\{ES_{j_1\omega}^O, \tau_{j_1}\},$$

which satisfies Eq. (5). Assume the Eq. (5) holds for the $(k-1)$ -th truck j_{k-1} in the sequence. We need to show that this holds for the k -th truck j_k . By definition, $\hat{s}_{j_k\omega}$ is given by

$$\begin{aligned} \hat{s}_{j_k\omega} &= \max\{\hat{a}_{j_k}^\omega, \tau_{j_k}, \min\{\hat{s}_{j_{k-1}\omega} + g_{j_{k-1}}, d_{j_{k-1}}\}\} \\ &= \max\{\hat{a}_{j_k}^\omega, \tau_{j_k}, \min\{\max\{ES_{j_{k-1}\omega}^O, \tau_{j_{k-1}}\} + g_{j_{k-1}}, d_{j_{k-1}}\}\} \end{aligned} \quad (\text{OS.1})$$

$$= \max\{\hat{a}_{j_k}^\omega, \tau_{j_k}, \min\{ES_{j_{k-1}\omega}^O + g_{j_{k-1}}, d_{j_{k-1}}\}, \min\{\tau_{j_{k-1}} + g_{j_{k-1}}, d_{j_{k-1}}\}\} \quad (\text{OS.2})$$

$$= \max\{ES_{j_k\omega}^O, \tau_{j_k}\} \quad (\text{OS.3})$$

Eq. (OS.1) follows from induction hypothesis. The equality in Eq. (OS.2) follows from the maximization. Eq. (OS.3) follows from definitions that $ES_{j_k\omega}^O = \max\{\hat{a}_{j_k}^\omega, \min\{ES_{j_{k-1}\omega}^O + g_{j_{k-1}}, d_{j_{k-1}}\}\}$ and $\tau_{j_k} \geq \min\{\tau_{j_{k-1}} + g_{j_{k-1}}, d_{j_{k-1}}\}$. By induction, the realized start time $\hat{s}_{j_k} = \max\{ES_{j_k\omega}^O, \tau_{j_k}\}$ holds for all trucks $j_k \in \pi_l$. □

The proof of Theorem 2

Proof. Let $\hat{s}_{j\omega}(\tau_j)$ be the realized start time under τ_j , we show that $\hat{s}_{j\omega}(\tau_j) = \hat{s}_{j\omega}(\tau'_j)$ holds for all $j \in \pi_l$. Proofs are by induction. Consider the first truck $j_1 \in \pi_l$, let $\tau'_{j_1} = \tau_{j_1}$, the result holds. Assume $\hat{s}_{j_{k-1}\omega}(\tau_{j_{k-1}}) = \hat{s}_{j_{k-1}\omega}(\tau'_{j_{k-1}})$ holds for $(k-1)$ -th truck j_{k-1} . Then for k -th truck j_k , by definition,

$$\begin{aligned} \hat{s}_{j_k\omega}(\tau_{j_k}) &= \max\{\hat{a}_{j_k}^\omega, \tau_{j_k}, \min\{\hat{s}_{j_{k-1}\omega}(\tau'_{j_{k-1}}) + g_{j_{k-1}\omega}, d_{j_{k-1}}\}\} \\ &= \max\{\hat{a}_{j_k}^\omega, \tau_{j_k}, \min\{ES_{j_{k-1}\omega}^O + g_{j_{k-1}}, d_{j_{k-1}}\}, \min\{\tau'_{j_{k-1}} + g_{j_{k-1}}, d_{j_{k-1}}\}\} \end{aligned} \quad (\text{OS.4})$$

$$= \max\{ES_{j_k\omega}^O, \tau'_{j_k}\} \quad (\text{OS.5})$$

$$= \hat{s}_{j_k\omega}(\tau'_{j_k}) \quad (\text{OS.6})$$

Eq. (OS.4) follows from Theorem 1. Eq. (OS.5) is derived from the definition of $ES_{j_k\omega}^O$ and $\tau'_{j_k} = \max\{\min\{\tau'_{j_{k-1}} + g_{j_{k-1}}, d_{j_{k-1}}\}, \tau_{j_k}\}$. $\square$

OS.2. Instance generation and algorithm parameters configuration

Table OS.1 Instance Generation Parameters

Parameter	Generation Method / Value	SetA	SetB	SetC
Non-zero flow value ($f_{ij} > 0$)	$U[6, 60]$ (Miao et al. 2014)	✓	✓	✓
Density	Fraction of non-zero flows	0.3	1.0	1.0
Penalty coefficient (p_{ij})	$U[15, 25]$	✓	✓	✓
Inbound arrival ($a_i, i \in \mathcal{I}$)	$U\left[1, \frac{60 \cdot \mathcal{I} }{ \mathcal{D}^I }\right]$ (Miao et al. 2014)	✓	✓	✓*
Outbound arrival ($a_j, j \in \mathcal{O}$)	$U\left[15, 15 + \frac{60 \cdot \mathcal{O} }{ \mathcal{D}^O }\right]$ (Miao et al. 2014)	✓	✓	✓*
Expected departure ($e_j, j \in \mathcal{I} \cup \mathcal{O}$)	$e_j = a_j + U[25, 40]$ (Miao et al. 2014)	✓	✓	✓
Latest departure ($d_j, j \in \mathcal{I} \cup \mathcal{O}$)	$d_j = e_j + U[10, 15]$	✓	✓	✓
Processing time ($g_j, j \in \mathcal{I} \cup \mathcal{O}$)	$U[5, 10]$	✓	✓	✓
Speed (v)	1	✓	✓	✓
Max load capacity (Q)	20	✓	✓	✓
Distance (h_{kl})	Manhattan distance between inbound door $k \in \mathcal{D}^I$ and outbound door $l \in \mathcal{D}^O$. Rectangular cross-dock layout with opposite doors fixed 3 units apart and adjacent doors on the same side 1 unit apart (Miao et al. 2009, 2014, Guignard et al. 2012).			

* SetC uses modified arrival times: $a_i \sim U[1, 2 \cdot |\mathcal{I}|]$, $a_j \sim U[15, 15 + 2 \cdot |\mathcal{O}|]$

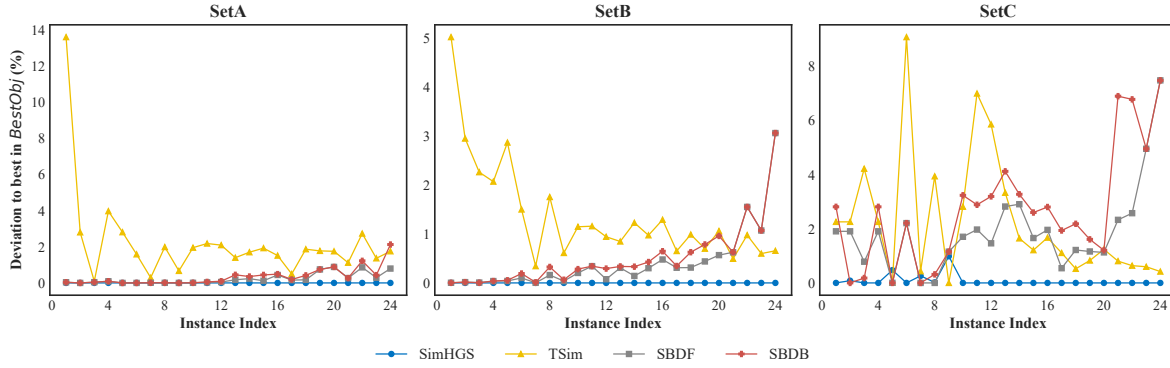
SimHGS uses the same parameter configuration for SetA and SetB instances and a different parameter configuration for SetC instances.

Table OS.2 Parameter configuration via Itrace (López-Ibáñez et al. 2016)

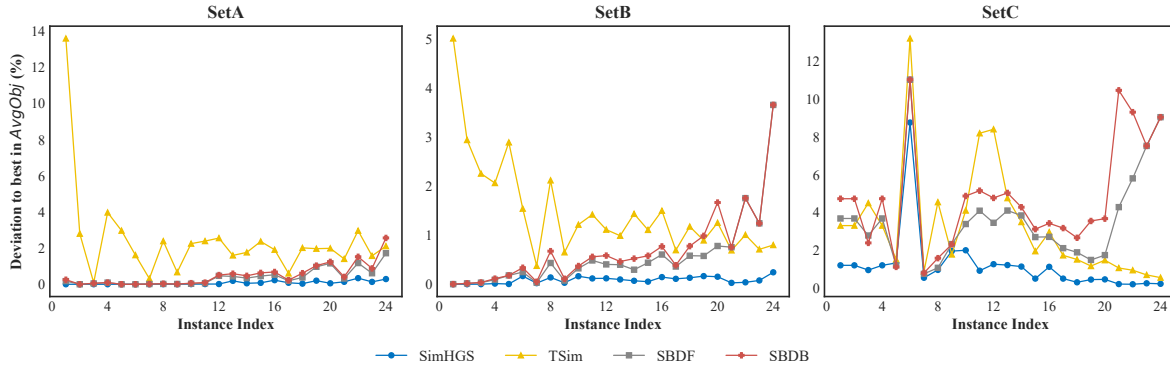
Param.	Section	Description	Tuning range	SetA & SetB	SetC
μ	3.6	Minimum population size	{5, 10, 15}	5	5
λ	3.6	Maximum increase in population size	{5, 10, 15, 20}	10	10
I_{max}	3.1	Maximum iteration of consecutive unimproved	{5000, 10000, 20000}	10000	5000
β	3.1	Tolerance to the best value for deterministic version	[0.6, 1.0]	1.0	0.9
n_{close}	3.4	Number of most similar individuals	{3, 4, 5, 6}	3	5
n_{elite}	3.4	Number of best elite individuals	{3, 4, 5, 6}	3	5

OS.3. Detailed comparison results of SimHGS against the reference algorithms

Figures OS.1a to OS.1b illustrate the deviations of each algorithm from the best known value, reporting both the best and average objective values across all instances. For three SetC instances (10x3C, 10x4C, 15x4C), TSim produced extreme deviations ($> 330\%$). To improve visualization, these outliers were replaced by the average deviation of the remaining instances. All detailed results are available at <https://github.com/yangwei-nwpu/SCDSP>.



(a) Best objective value



(b) Average objective value

Figure OS.1 Deviations of the best and average objective values from the reference results

OS.4. The deterministic MILP model for SCDSP

Table OS.3 Variables for Cross-Dock Scheduling Problem

Notation	Description
x_{ik}	1 if inbound truck $i \in \mathcal{I}$ is assigned to unloading door $k \in \mathcal{D}^{\mathcal{I}}$, 0 otherwise
y_{jl}	1 if outbound truck $j \in \mathcal{O}$ is assigned to loading door $l \in \mathcal{D}^{\mathcal{O}}$, 0 otherwise
z_{ijkl}	1 if goods flow f_{ij} is successfully transferred via inbound door $k \in \mathcal{D}^{\mathcal{I}}$ to outbound door $l \in \mathcal{D}^{\mathcal{O}}$, 0 otherwise
s_i	Processing start time of truck $i \in \mathcal{I} \cup \mathcal{O}$
$\zeta_{ii'}$	1 if truck i, i' are assigned to the same door and i is processed before i' , 0 otherwise
t_i	Departure tardiness of truck $i \in \mathcal{I} \cup \mathcal{O}$

To further evaluate the effectiveness of the HGS algorithm, we consider the deterministic version of the SCDSP and formulate it as a mixed-integer linear programming (MILP) model. This mathematical formula-

tion serves as an exact benchmark for assessing both the solution quality and computational efficiency of our heuristic. The notation and decision variables employed in the MILP model are summarized in Table OS.3.

$$\text{MILP min } c \sum_{i \in \mathcal{I} \cup \mathcal{O}} t_i + \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{O}} p_{ij} f_{ij} \left(1 - \sum_{k \in \mathcal{D}^{\mathcal{I}}} \sum_{l \in \mathcal{D}^{\mathcal{O}}} z_{ijkl} \right) \quad (\text{OS.7})$$

$$\text{s.t. } \sum_{k \in \mathcal{D}^{\mathcal{I}}} x_{ik} \leq 1, \quad \forall i \in \mathcal{I} \quad (\text{OS.8})$$

$$\sum_{l \in \mathcal{D}^{\mathcal{O}}} y_{jl} \leq 1, \quad \forall j \in \mathcal{O} \quad (\text{OS.9})$$

$$\zeta_{ii'} + \zeta_{i'i} \geq x_{ik} + x_{i'k} - 1, \quad \forall (i, i') \in \mathcal{I}^2, i < i', \forall k \in \mathcal{D}^{\mathcal{I}} \quad (\text{OS.10})$$

$$\zeta_{jj'} + \zeta_{j'j} \geq y_{jl} + y_{j'l} - 1, \quad \forall (j, j') \in \mathcal{O}^2, j < j', \forall l \in \mathcal{D}^{\mathcal{O}} \quad (\text{OS.11})$$

$$\zeta_{ii'} + \zeta_{i'i} \leq 1, \quad \forall (i, i') \in \mathcal{I}^2 \cup \mathcal{O}^2, i < i' \quad (\text{OS.12})$$

$$s_{i'} - s_i - (M + g_i) \zeta_{ii'} \geq -M, \quad \forall (i, i') \in \mathcal{I}^2 \cup \mathcal{O}^2, i \neq i' \quad (\text{OS.13})$$

$$z_{ijkl} \leq x_{ik}, \quad \forall i \in \mathcal{I}, j \in \mathcal{O}, k \in \mathcal{D}^{\mathcal{I}}, l \in \mathcal{D}^{\mathcal{O}} \quad (\text{OS.14})$$

$$z_{ijkl} \leq y_{jl}, \quad \forall i \in \mathcal{I}, j \in \mathcal{O}, k \in \mathcal{D}^{\mathcal{I}}, l \in \mathcal{D}^{\mathcal{O}} \quad (\text{OS.15})$$

$$s_j - s_i - (M + q_{ij} h_{kl} + g_i) z_{ijkl} \geq -M, \quad \forall i \in \mathcal{I}, j \in \mathcal{O}, k \in \mathcal{D}^{\mathcal{I}}, l \in \mathcal{D}^{\mathcal{O}} \quad (\text{OS.16})$$

$$t_i \geq s_i + g_i - e_i, \quad \forall i \in \mathcal{I} \cup \mathcal{O} \quad (\text{OS.17})$$

$$s_i \geq a_i, \quad \forall i \in \mathcal{I} \cup \mathcal{O} \quad (\text{OS.18})$$

$$s_i \leq d_i - g_i, \quad \forall i \in \mathcal{I} \cup \mathcal{O} \quad (\text{OS.19})$$

$$x_{ik} \in \{0, 1\}, \quad \forall i \in \mathcal{I}, k \in \mathcal{D}^{\mathcal{I}} \quad (\text{OS.20})$$

$$y_{jl} \in \{0, 1\}, \quad \forall j \in \mathcal{O}, l \in \mathcal{D}^{\mathcal{O}} \quad (\text{OS.21})$$

$$\zeta_{ii'} \in \{0, 1\}, \quad \forall (i, i') \in \mathcal{I}^2 \cup \mathcal{O}^2, i \neq i' \quad (\text{OS.22})$$

$$z_{ijkl} \in \{0, 1\}, \quad \forall i \in \mathcal{I}, j \in \mathcal{O}, k \in \mathcal{D}^{\mathcal{I}}, l \in \mathcal{D}^{\mathcal{O}} \quad (\text{OS.23})$$

$$t_i \geq 0, \quad \forall i \in \mathcal{I} \cup \mathcal{O} \quad (\text{OS.24})$$

The objective function (OS.7) minimizes the total cost comprising truck tardiness and penalties for untransferred goods. Constraints (OS.8) and (OS.9) assign each inbound and outbound truck to at most one door. Constraints (OS.10)–(OS.13) govern the sequencing of trucks assigned to the same door and ensure nonoverlapping processing. Constraints (OS.14)–(OS.16) link goods transfer decisions to door assignments and timing feasibility. Constraints (OS.17)–(OS.19) define tardiness and enforce arrival time and deadline restrictions. Constraints (OS.20)–(OS.24) specify the variable domains.

Table OS.4 presents a comparative analysis of the proposed HGS algorithm and the MILP model across three benchmark sets (SetA, SetB, and SetC). For HGS, we report the best and average objective values from 10 independent runs, alongside the average runtime (limited to 600 seconds per run). The MILP model was solved using CPLEX 22.10 with a 3,600 seconds time limit. Bold values indicate the better result and optimal solutions obtained by the MILP model are highlighted with an asterisk “*”. The results demonstrate that the HGS algorithm consistently outperforms the MILP model across all benchmark sets in both efficiency and solution quality. For smaller instances where the MILP model achieves proven optimality, HGS identifies the same optimal solutions in significantly less computational time. In more complex cases

Table OS.4 Detailed comparison results between HGS algorithm and MILP model across all instances

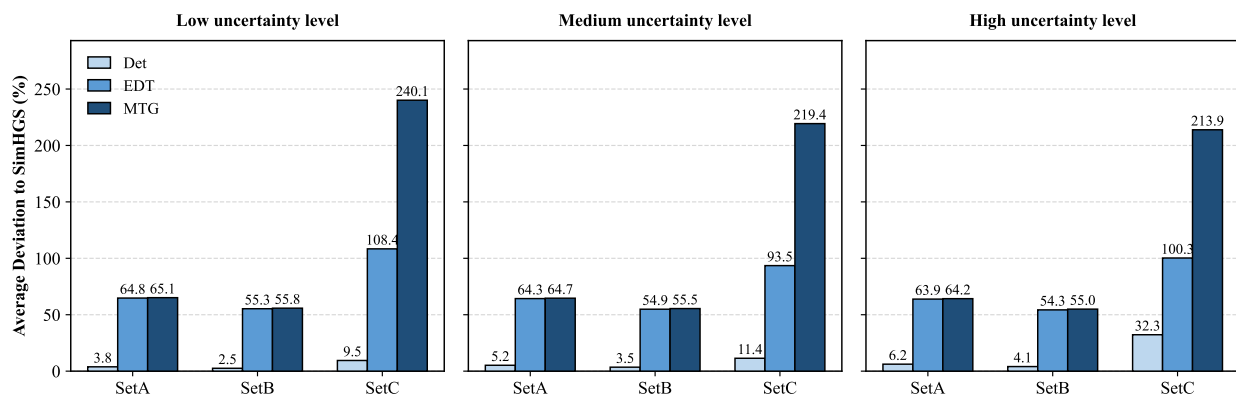
Inst.	Set A					Set B					Set C				
	HGS			MILP		HGS			MILP		HGS			MILP	
	Best	AvgObj	AvgT	Best	Time	Best	AvgObj	AvgT	Best	Time	Best	AvgObj	AvgT	Best	Time
10x3	5506	5506	0.1	5506*	0.9	17388	17388	0.3	17388*	69	4	5.4	246	4*	2861
10x4	7323	7323	0.0	7323*	109	23637	23637	0.3	23637*	584	0	0	82	0*	65
15x3	10014	10014	0.3	10014*	4.9	37643	37643	1.1	37643	3590	5004	5004	67	8305	3590
15x4	13198	13198	0.4	13198*	209	50171	50171	1.1	50171	3584	67	70.2	277	3057	3591
20x3	17683	17683	1.5	17683*	18	62015	62015	73	62015	3585	23316	23494	207	39982	3580
20x4	17132	17132	2.1	17132*	1260	66644	66644	6.1	66648	3591	1129	1232	102	17639	3591
25x3	70383	70383	27	70383*	150	241587	241587	122	241588	3590	69462	70075	238	94796	3589
25x5	35517	35517	5.0	35517	3566	115114	115116	418	115142	3589	38012	38214	482	60446	3590
28x3	64508	64508	18	64508*	67	240743	240743	84	240743	3586	105510	106402	375	147167	3590
28x5	62791	62791	12	62791	3580	195792	195792	144	195808	3591	36884	36896	499	125709	3591
30x6	52850	52850	92	52866	3586	193411	193412	336	195768	3584	30898	30901	321	56571	3590
30x8	61223	61223	70	62123	3590	198136	198138	274	208295	3591	30874	30881	452	397378	3590
35x6	88426	88427	149	88427	3589	304140	304142	360	308808	3591	103916	104440	466	184767	3585
35x8	83009	83009	226	83024	3590	263681	263684	340	274114	3585	99385	99759	452	523792	3591
40x6	106645	106646	425	107496	3590	345278	345285	416	352098	3586	165861	166771	445	254085	3590
40x8	118276	118282	449	118911	3590	359056	359059	536	376610	3590	163852	164889	318	248570	3586
50x6	259785	259787	381	259805	3591	830089	830098	446	836103	3592	399607	400781	494	1487770	3592
50x8	167080	167084	483	170956	3591	522623	522632	504	767383	3590	389848	391233	361	1441470	3592
60x8	255983	255999	492	265100	3587	934177	934202	394	1161790	3607	504997	505783	326	2179280	3591
60x10	255826	255838	523	266034	3591	845184	845213	395	2359840	3619	504312	506394	398	2359850	3612
80x8	551678	551707	527	722316	3593	1808850	1809117	368	4221240	3621	1053370	1056179	328	4221250	3623
80x10	369103	369140	424	392307	3593	1319790	1320270	424	4221250	3624	1055420	1057383	382	4221250	3599
100x10	766271	766329	371	1941110	3619	2595660	2597305	400	6608060	3628	2168350	2171574	280	6608070	3602
100x15	739248	739312	342	1941130	3608	2490820	2493656	369	6608060	3654	2175300	2179649	368	6608070	3639

where the MILP model fails to converge within the time limit, HGS consistently produces superior objective values. These findings highlight that HGS consistently outperforms the MILP model, delivering superior solution quality with significantly higher computational efficiency for large-scale instances.

OS.5. Sensitivity analysis

This section examines the robustness of the proposed SimHGS and the impact of key parameters. Section OS.5.1 analyzes the performance of SimHGS under different uncertainty levels. Section OS.5.2 investigates the trade-off between truck tardiness and untransferred goods by varying the unit tardiness cost. Section OS.5.3 studies the effect of scenario size on neighborhood evaluation accuracy.

OS.5.1. Sensitivity analysis of uncertainty levels

**Figure OS.2 Average deviations of baseline policies from SimHGS under different uncertainty levels.**

To ensure that our findings are not contingent on a specific arrival-time distribution, we evaluate the baseline policies across three distinct uncertainty levels: low ($Tri(-3, 3, 0)$), medium ($Tri(-6, 6, 0)$), and high ($Tri(-9, 9, 0)$). Figure OS.2 illustrates the average deviations of Det, EDT, and MTG relative to SimHGS across SetA, SetB, and SetC. The results indicate that the relative performance ranking of all methods remains remarkably stable across all tested uncertainty levels. Specifically, SimHGS consistently outperforms Det, EDT, and MTG in every scenario. This superiority is most pronounced in SetC, where EDT and MTG exhibit substantial performance gaps compared to SimHGS. Notably, the performance of the Det policy deteriorates significantly as uncertainty increases, whereas SimHGS maintains its lead. These observations confirm that the advantages of SimHGS are robust to variations in arrival-time distribution parameterization, reinforcing its reliability in diverse uncertain environments.

OS.5.2. Sensitivity analysis of objective weights

To evaluate the trade-off between service quality (truck tardiness) and throughput (untransferred goods), we conduct a sensitivity analysis on the unit tardiness cost c . Using the SimHGS algorithm, we vary $c \in \{1, 10, 20, 30, 40, 50\}$ across three representative instances (30x6A, 30x6B, 30x6C). As shown in Figure OS.3, the left y-axis represents the raw average total tardiness in time units rather than the weighted cost ($c \times$ tardiness). This choice provides a direct and intuitive assessment of operational performance. By excluding the scaling effect of c , we avoid the mathematical inflation of objective values that occurs as the weight increases, thereby providing a clearer visualization of the actual reduction in physical delays.

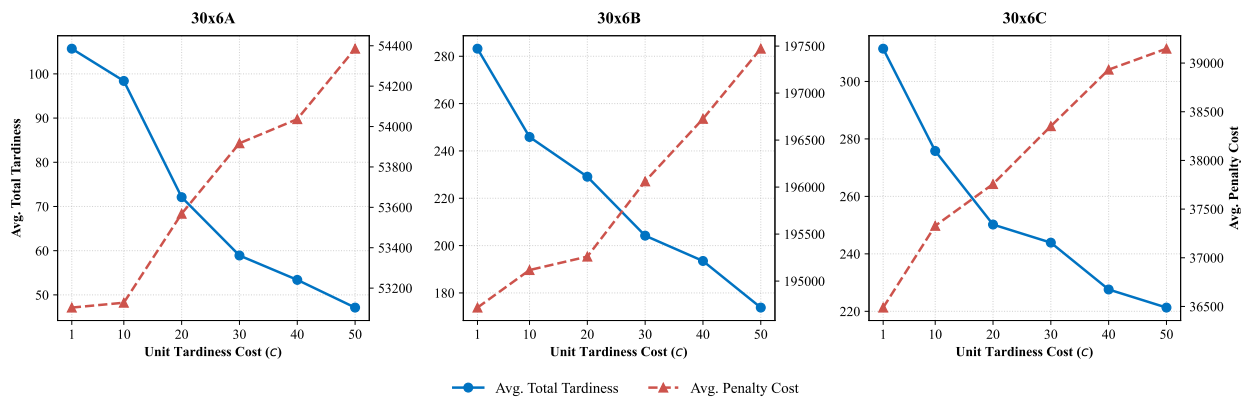


Figure OS.3 The unit tardiness cost analysis for different c

As illustrated in Figure OS.3, a distinct trade-off emerges between the two competing objectives. As the unit tardiness cost c increases, the objective function prioritizes adherence to scheduled departure times, resulting in a monotonic reduction in average tardiness (represented by solid blue lines). However, this stricter enforcement of schedule punctuality comes at the expense of throughput, as evidenced by the concomitant rise in penalties for untransferred goods (dashed red lines). These results demonstrate that SimHGS is highly sensitive to management priorities, allowing for a seamless calibration between strict schedule adherence and maximized throughput. This adaptability confirms the model's effectiveness in supporting decision-making under diverse cost structures and operational strategies.

OS.5.3. Sensitivity analysis of scenario size

To determine the scenario size used in SBLS, we conducted a sensitivity analysis on neighborhood evaluation with $N_s \in \{100, 500, 1000, 1500\}$. For each N_s , the best move identified during a neighborhood search iteration was compared with a benchmark move obtained under a reference evaluation with $N_{ref} = 2000$ scenarios. A misclassification was recorded whenever the two evaluations yielded different best moves. The resulting loss was quantified by the regret (the difference in objective values under the reference evaluation) and its corresponding percentage deviation. For each instance, these metrics were averaged over all neighborhood search iterations during the descent process across 10 independent runs. As illustrated in Figure OS.4, the percentage deviation decreases as N_s increases across all benchmark sets. The reduction is pronounced when N_s increases from 100 to 1000; however, the marginal improvement from 1000 to 1500 is notably smaller. These observations indicate diminishing marginal returns when the scenario size exceeds 1000, justifying the selection of $N_s = 1000$ as an effective balance between evaluation accuracy and computational overhead.

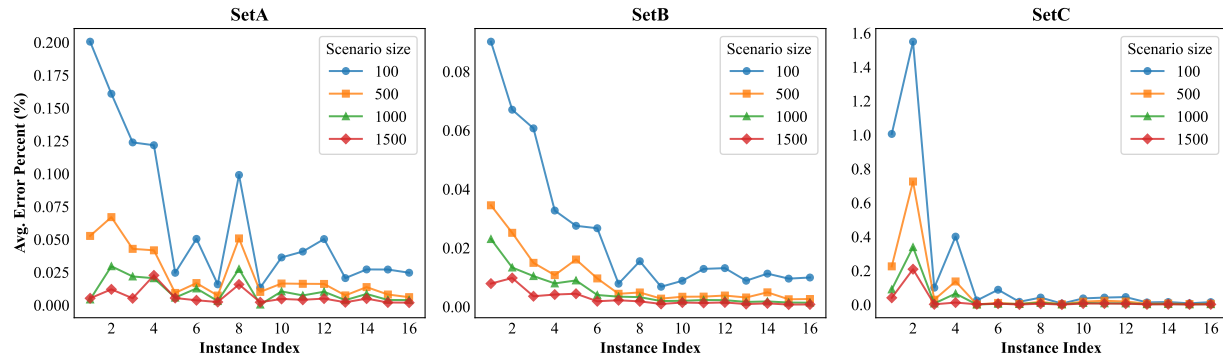
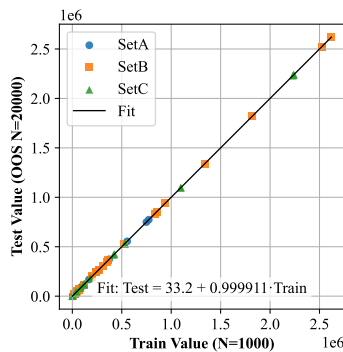
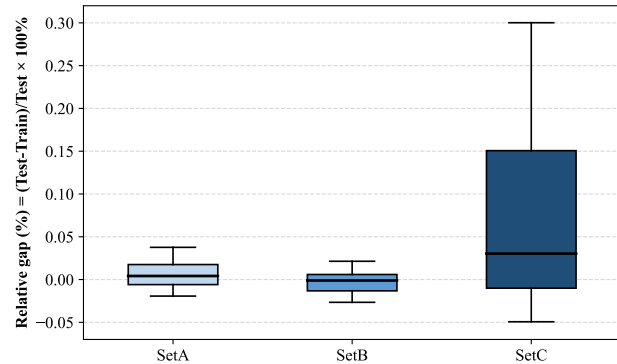


Figure OS.4 The average error percentage for different scenario size across all instances



(a) Consistency between in-sample and out-of-sample objective values



(b) Distribution of relative gaps across instance sets (SetA–SetC)

Figure OS.5 Out-of-sample ($N = 20000$) validation of the scenario-based evaluation ($N = 1000$)

To assess whether $N_s = 1000$ ensures robust out-of-sample performance, we evaluated the resulting scheduling policies against an independent test pool of 20,000 scenarios. Figure OS.5a reveals a near-perfect

linear correlation ($\text{Test} \approx 0.9999 \times \text{Train}$), suggesting that the in-sample evaluations are unbiased estimators of the true objective values. Furthermore, Figure OS.5b further shows a close alignment between the two values within a remarkably tight margin of $\pm 0.05\%$. Collectively, these results justify the use of $N_s = 1000$ in SBLS as an optimal equilibrium between statistical precision and computational tractability.

OS.6. Component effectiveness of SimHGS

This section analyzes the contribution of the main components of SimHGS. Section OS.6.1 provides a runtime breakdown and evaluates the efficiency of the fast evaluation technique. Section OS.6.2 assesses the effectiveness of the cutoff rule. Section OS.6.3 quantifies the contribution of the simulation-based local search. Section OS.6.4 evaluates the role of the hybrid genetic search framework.

OS.6.1. Runtime breakdown and efficiency analysis

To provide quantitative evidence for the efficacy of the fast evaluation technique, Figure OS.6 illustrates the runtime decomposition of SimHGS across all 72 instances, categorized into VND, deterministic LS, and SBLS. Additionally, it reports the average percentage of affected scenarios (Aff.Scen.) requiring updates during the fast evaluation process.

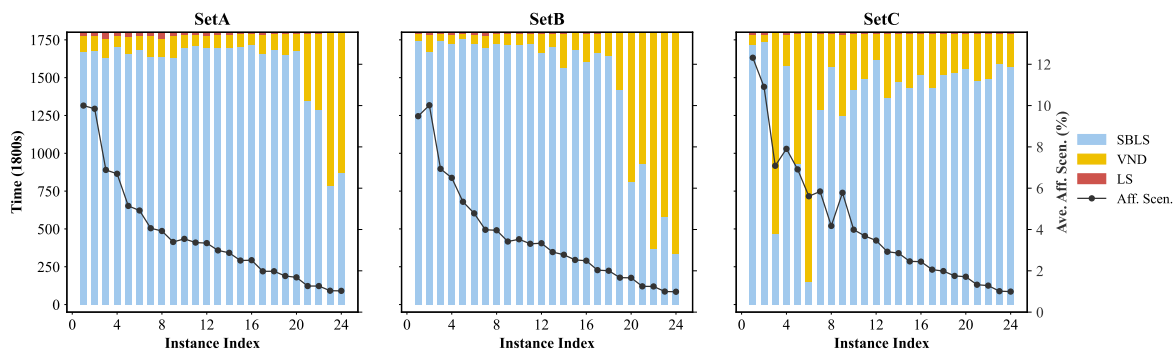


Figure OS.6 SimHGS runtime breakdown and average percentage of affected scenarios across all instances.

As shown in Figure OS.6, the proportion of affected scenarios remains consistently low across all instances, typically ranging from 1% to 12%. This confirms that only a small fraction of the scenario set necessitates re-evaluation after each timing move. This empirical observation aligns with the results presented in Section 4.4.2 and explains why the proposed technique achieves a speedup of 1–2 orders of magnitude over full evaluation, as reported in Section 6.5.

Figure OS.6 further elucidates the computational bottlenecks of SimHGS. For SetA and SetB, the computational effort dedicated to VND generally scales with problem size, suggesting that sequence optimization becomes the primary bottleneck for larger instances. In contrast, for SetC, the deterministic LS accounts for a disproportionately larger share of the total runtime (averaging 12 seconds for 100-truck instances in SetC, versus 4 seconds and 2 seconds for SetA and SetB). This indicates that the cutoff rule is less restrictive on SetC, allowing more candidate sequences to bypass the initial filter and enter the local improvement

phase. The resulting increase in LS and SBLS invocations elevates the overall computational burden for this set. Collectively, these results complement the theoretical and empirical analyses in Sections 4.4.2 and 6.5, providing a granular understanding of how the fast evaluation technique mitigates the simulation-based computational overhead.

OS.6.2. Effectiveness of the cutoff rule

In the SimHGS algorithm, only solutions that satisfy the cutoff rule (promising deterministic solutions π) trigger the optimization of the corresponding release times τ . This selective strategy enhances the overall search performance. To verify the effectiveness of the cutoff rule, we compare SimHGS with NoCutoff, a variant that disables the cutoff rule.

Table OS.5 Summary of comparison results between SimHGS and NoCutoff

Instances	<i>BestObj</i>			<i>AvgObj</i>		
	#Wins	#Ties	#Losses	#Wins	#Ties	#Losses
SetA	13	8	3	18	5	1
SetB	20	3	1	19	2	3
SetC	14	2	8	21	0	3
Total	47	13	12	58	7	7
<i>p</i> -value	2.02E-06			5.14E-10		

Table OS.5 summarizes the comparison between the SimHGS and NoCutoff variants in terms of best and average objective values. The analysis shows that SimHGS outperforms NoCutoff for 47 out of 72 instances for the best objective value, while performing worse for 12 instances. For the average objective value, SimHGS achieves better results for 58 instances and worse for 7 instances. Interestingly, most of the instances where SimHGS underperforms are small instances with no more than 50 trucks. This suggests that although the cutoff rule may occasionally exclude some promising solutions, its advantages in improving search efficiency for large instances clearly outweigh these limitations. The benefits are especially pronounced in the more challenging SetB and SetC, where SimHGS consistently delivers stronger performance. Overall, the cutoff rule proves effective in filtering out unpromising truck sequences π , thereby boosting the algorithm's overall performance across all instance types.

OS.6.3. Effectiveness of the SBLS

To quantify the contribution of SBLS beyond deterministic scheduling, we conducted a paired comparison using the final best policies identified by SimHGS. Let $\Pi^* = (\pi, \tau^*)$ denote the best policy after SBLS refinement, and let $\Pi = (\pi, \tau)$ represent its deterministic predecessor, i.e., the policy immediately prior to the application of SBLS. By construction, both policies share an identical truck sequence π and differ exclusively in their timing decisions. Both Π^* and Π were evaluated on the same training set of 1000 scenarios. The relative improvement attributable to SBLS over the deterministic local search (LS) is calculated as $gap = \frac{LS - SBLS}{LS} \times 100\%$. As illustrated in Figure OS.7, SBLS consistently yields positive improvements

across all benchmark sets, with average gains of 4.47% on SetA, 2.75% on SetB, and 15.05% on SetC. Since the sequence π is unchanged, these performance gains are entirely driven by the stochastic refinement of timing decisions. This demonstrates that SBLS enhances the quality of the final SimHGS solutions by effectively adapting deterministic schedules to accommodate stochastic arrival time uncertainties.

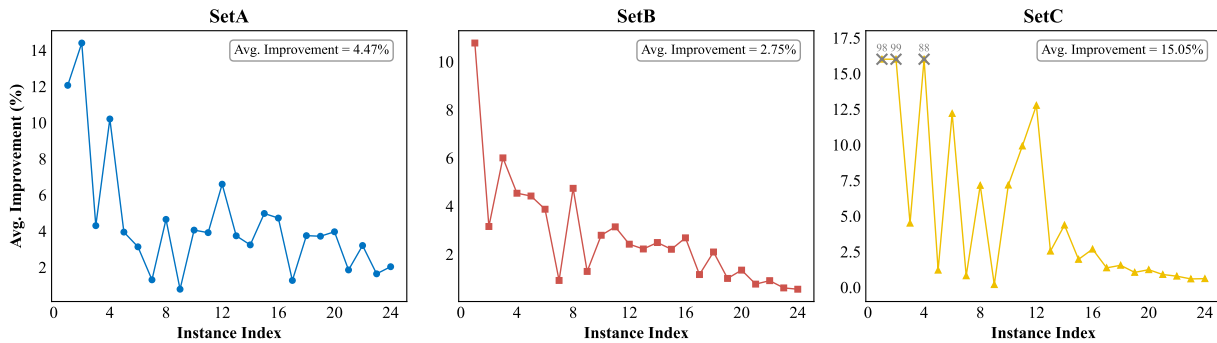


Figure OS.7 The average improvement of SBLS over LS across all instances

OS.6.4. Effectiveness of the HGS

To evaluate the contribution of the hybrid genetic search (HGS), we compare SimHGS with MS-NSBLS, a multi-start variant that repeatedly applies the nested simulation-based local search (NSBLS) from diverse initial solutions. Both algorithms are executed under a strict time limit of 1,800 seconds. This comparison is designed to distinguish the benefits of the population-based evolutionary mechanism from those achieved purely through independent local search iterations.

The summarized results in Table OS.6 reveal a consistent and statistically significant superiority of SimHGS across all benchmarks. For the best objective value, SimHGS outperforms MS-NSBLS for 57 instances, ties for 12 instances, and loses for 3 instances, yielding a highly significant p -value of 1.16E-10. The performance gap is similarly pronounced for the average objective value, where SimHGS wins for 60 instances, ties for 8 instances, and loses for 4 instances (p -value = 4.00E-11). These findings demonstrate that SimHGS's superior performance is not merely due to multiple restarts. Instead, the population-based mechanism effectively maintains diversity and facilitates information exchange between solutions. This synergy allows the algorithm to explore promising regions of the solution space that remain inaccessible to isolated local searches, thereby confirming HGS as an essential component of our methodology.

Table OS.6 Summary of comparison results between SimHGS and MS-NSBLS

Instances	<i>BestObj</i>			<i>AvgObj</i>		
	#Wins	#Ties	#Losses	#Wins	#Ties	#Losses
SetA	15	6	3	16	5	3
SetB	22	2	0	21	2	1
SetC	20	4	0	23	1	0
Total	57	12	3	60	8	4
p -value	1.16E-10			4.00E-11		